REVIEW

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Commemorating the National Institute of Diabetes and Digestive and Kidney Diseases' Advances in Kidney Health

75 Years of Discovery and Impact

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Abstract

This year commemorates the 75th anniversary of the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), one of the 27 institutes and centers of the National Institutes of Health. A core mission of the NIDDK has been the advancement and support of biomedical research across a diverse spectrum of disciplines, including endocrine and metabolic diseases, digestive and nutritional disorders, obesity, urologic and benign hematologic conditions, and, notably, kidney diseases, which has been a major focus of the institute's strategic priorities. Through the years, the NIDDK has heavily invested in biomedical infrastructure, foundational studies, and cross-cutting basic science, clinical investigation, epidemiology, and health services research, which have fundamentally shaped the detection, management, and prevention of kidney diseases worldwide. Furthermore, the NIDDK has had a longstanding commitment to promoting workforce development, advancing equal access to kidney health care, and forging collaborative partnerships with academic centers, federal agencies, professional societies, patient advocacy organizations, community groups, and industry stakeholders toward the shared goal of improving kidney disease outcomes. In this review published across the three American Society of Nephrology journals, we celebrate the landmark achievements and profound effect of the NIDDK in improving the health and well-being of people living with kidney diseases worldwide.

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Introduction

Over the course of its 75-year history, the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) has had a pivotal role in advancing kidney research, inspiring scientific discovery, elevating standards of care, promoting health, and cultivating the next generation of scientists and clinicians. As an institute of the National Institutes of Health (NIH), one of the NIDDK's missions has been to conduct and support medical research across a broad range of areas—including endocrine and metabolic, digestive, nutritional, obesity, urologic, benign hematologic, and kidney disorders—with kidney-related research as a core pillar of its strategic priorities (Figure 1).[1] Through the years, the NIDDK's objectives have continuously evolved to address the changing landscape of kidney disease research and policy. Although in its earlier years, the institute particularly influenced the development, refinement, and implementation of treatments for kidney failure, over time, it further expanded its priorities to encompass CKD, AKI, glomerular diseases, mechanisms of kidney injury, hypertension, genetic diseases, prevention strategies, and health disparities. This progression underscores NIDDK's visionary leadership in fostering cross-cutting comprehensive approaches that integrate basic science, clinical investigation, epidemiology, and health services research to improve outcomes for patients across the spectrum of kidney diseases.

In honor of this year's NIDDK's 75th anniversary, this review published across the three American Society of Nephrology (ASN) journals celebrates the enduring legacy of the institute while critically reflecting on its profound scientific, clinical, and public health impact in combating kidney diseases, one of the most prevalent, fatal, and costly conditions worldwide. CKD affects 35.5 million people in the United States (14% of the US adult population)[2] and 850 million globally. Kidney failure treated with KRT

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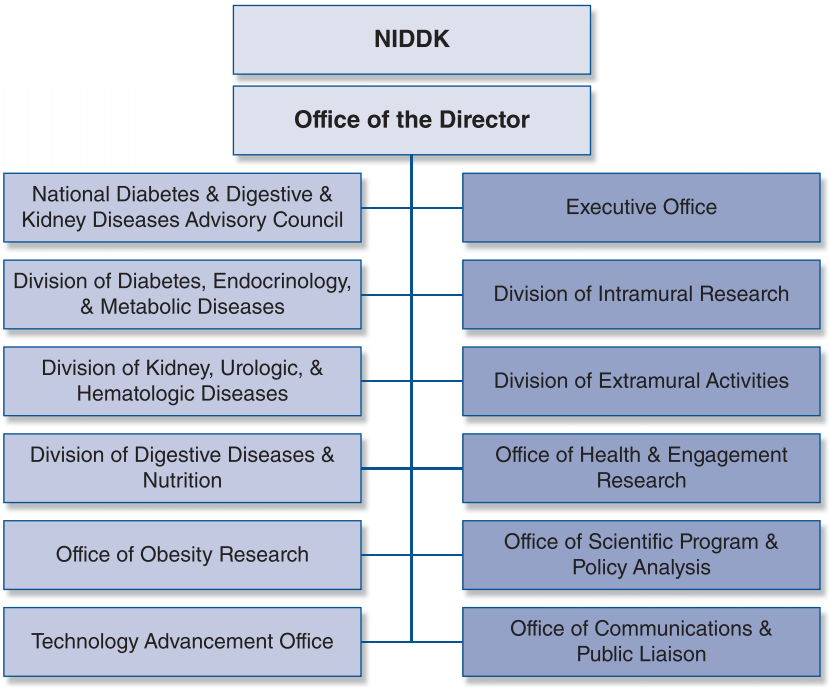

Figure 1. NIDDK's organizational structure. Adapted from https://www.niddk.nih.gov/about-niddk/offices-divisions/organizational-chart. NIDDK, National Institute of Diabetes and Digestive and Kidney Diseases.

(dialysis or transplantation) affects over 880,000 people in the United States.[3] In the United States alone, approximately 360 people initiate KRT on a daily basis (*i.e.*, approximately 130,000 individuals annually progressing to kidney failure), with those initiating dialysis having a 7- to 10-fold higher death risk compared with their non-CKD counterparts. Furthermore, total annual Medicare Fee for Service (FFS) spending for beneficiaries with CKD was $95.7 billion (*i.e.*, 25% of total Medicare FFS expenditures), with an additional $30.8 billion spent for beneficiaries with kidney failure in 2022.[4] Population health data also show that the burden of AKI has increased over time, including AKI-related hospitalizations and dialysis-requiring AKI.[4,5] For example, in the United States, the percentage of hospitalizations in which AKI was diagnosed among older adults receiving Medicare FFS or Medicare Advantage increased from 17% in 2012 to 26% in 2022.[4]

Although this review largely focuses on the contributions of the NIDDK, it should be acknowledged that other NIH institutes have also provided longstanding and significant investment into kidney-related research. While a review of their contributions is beyond the scope of this article, the ongoing investment of these collective institutes in eradicating kidney diseases and improving kidney health and its complications has been transformative to the field. By highlighting some of the NIDDK's major scientific breakthroughs, landmark studies, policy advances, and commitment to training the next generation of kidney health leaders, this review aims to commemorate NIDDK's impactful contributions and commitment to improving the health and well-being of people living with kidney diseases.

Early Years: Building the Foundation for Kidney Research

The early beginnings of NIDDK's kidney research activities traces back to the mid-20th century, a period characterized by rapid advances in biomedical science and growing recognition of chronic diseases as a major public health issue.[1,6] Originally established in 1950 as the National Institute of Arthritis and Metabolic Diseases, the institute was tasked with addressing a broad spectrum of metabolic disorders, including those affecting kidney function. The post–World War II era saw an expansion of federal investment in biomedical research, driven by societal needs to treat chronic illnesses that increasingly affected the aging population. This initial period laid the groundwork for investigations of mechanisms of kidney diseases and therapies. Soon after its establishment as the National Institute of Arthritis and Metabolic Diseases, the institute began funding studies in kidney physiology. Trailblazers in kidney physiology such as Dr. Homer Smith advanced key insights into tubular function, noninvasive measurement of glomerular filtration, and mechanisms of water excretion and electrolyte balance, providing a foundation for contemporary nephrology.[7,8] During the 1960s, discoveries advanced by NIDDK-supported research established proteinuria as a marker of glomerular injury and serum creatinine as a practical measure of kidney function. Concurrently, pioneering efforts led by Dr. Willem Kolff and others to transition dialysis from an experimental therapy to a life-saving intervention for kidney failure were accelerated with NIDDK support.[9] By the late 1960s, hemodialysis became increasingly accessible, dramatically improving survival rates for patients with kidney

failure. To advance kidney research, the NIDDK strategically expanded both extramural and intramural programs. In the early 1970s, a hallmark achievement was the establishment of the George M. O'Brien Kidney Research Core Centers, creating multidisciplinary hubs that fostered innovation, collaboration, and training, and which continue to support kidney research today.[1,6] These centers provided critical infrastructure to support cutting-edge basic and translational research. Concurrently, NIDDK's intramural programs developed robust nephrology research efforts, contributing to a growing understanding of kidney physiology and pathology.

In 1986, reflecting an expanding research mission, the institute underwent a significant reorganization and was renamed the NIDDK.[1,6] This change formally acknowledged the inclusion of diabetes, digestive, kidney, and benign hematologic diseases under its purview. This realignment also underscored the institute's commitment to integrated approaches addressing complex, inter-related disorders and positioned NIDDK to lead comprehensive kidney disease research efforts with enhanced focus and resources. The NIDDK also played a crucial role in shaping national kidney health policy and key legislative milestones even during its early decades. For example, the 1972 Social Security Amendments, which extended Medicare coverage for KRT to patients with kidney failure, effectively creating the Medicare End-Stage Renal Disease Program,[10] were influenced by NIDDK-funded data. This milestone transformed kidney disease treatment by increasing access to life-saving dialysis and kidney transplantation services. Beyond legislation, the NIDDK has spearheaded the development and dissemination of national kidney health reports such as the United States Renal Data System (USRDS) Annual Data Reports[4,11,12] and strategic plans that have guided research priorities and public health initiatives, reinforcing the institute as a driving force of kidney disease practice and policy.

Pioneering Discoveries and Scientific Breakthroughs

Over the past seven and a half decades, research supported by the NIDDK has been fundamental to advancing the science and clinical care of kidney diseases. The institute's investments have catalyzed foundational discoveries, landmark clinical trials, and novel technologies that continue to shape nephrology, with some of its major milestones chronicled below (Figure 2).

Foundational Discoveries by the NIDDK

The NIDDK has been pivotal in advancing foundational bench research discoveries that, while often unfolding over many years, have been essential to the development of transformative clinical advances. For example, seminal studies led by Dr. Ernest Wright that identified the sodium-glucose cotransporter 2 (SGLT2) transporter, elucidated mechanisms of sodium and glucose transport, and ultimately led to the development of SGLT2 inhibitors were supported by the NIDDK.[13–16] NIDDK-funded research also demonstrated the profibrotic effects of aldosterone in preclinical models,[17–20] providing critical mechanistic insights leading to the development of a Food and Drug Administration (FDA)–approved mineralocorticoid receptor antagonist (*i.e.*, finerenone) that inhibits aldosterone signaling and diabetic kidney disease (DKD) progression.[21] In kidney physiology and transport, NIDDK-funded basic research has advanced understanding of how low dietary potassium and high-sodium intake leads to hypertension,[22] informing clinical studies in this arena. In addition, NIDDK-supported research also led to the identification and cloning of the calcium-sensing receptor,[23] leading to the development of calcimimetics used in mineral bone disease management. In addition to clarifying vasopressin's role in kidney tubular water handling, NIDDK-supported preclinical studies also demonstrated that vasopressin promotes polycystic kidney disease (PKD) progression.[24–32] These findings led to the development of tolvaptan, a vasopressin V2 receptor antagonist and the only FDA-approved therapy for slowing PKD progression.[33,34] NIDDK-funded basic research has also been instrumental in advancing our understanding of podocyte biology, function, and therapeutic interventions for nephrotic syndrome.[35–38] This includes NIDDK-supported basic research demonstrating the antiproteinuric effects of calcineurin inhibitors through stabilization of synaptopodin and preservation of podocyte integrity.[39,40] NIDDK-supported fundamental research also established the foundational scientific framework enabling the xenotransplantation of porcine kidneys into human recipients and for which industry trials are ongoing.[41,42]

CKD

NIDDK-supported research has been instrumental in shaping the understanding, classification, and treatment of CKD, from its pathogenesis to its progression and its related complications. For example, the Modification of Diet in Renal Disease (MDRD) study (1989–1994) was a landmark clinical trial that evaluated the effects of dietary protein restriction and BP control in slowing CKD progression.[43] The trial also led to the development of the MDRD equation,[44] one of the first validated methods to estimate GFR, which was pivotal in shifting the field from relying solely on serum creatinine to measure kidney function and providing a framework for standardized CKD staging. In 2009, the CKD-Epidemiology Collaboration (CKD-EPI) equation was developed to improve the accuracy of eGFR calculations, particularly at higher GFR levels (*i.e.*, above 60 ml/min per 1.73 m^2), compared with the MDRD equation.[45] Given growing recognition of the problems with the use of race in the equation, in 2022, a Task Force convened by the ASN and National Kidney Foundation recommended the use of a new CKD-EPI equation for eGFR that removes the race coefficient, as well as utilization of cystatin C in conjunction with serum creatinine to further minimize population-specific biases and improve accuracy.[46,47]

Recognizing the high prevalence of kidney diseases among African American individuals, the African American Study of Kidney Disease and Hypertension (AASK) trial (1995–2001) was the first large randomized controlled trial in CKD specifically focused on participants with a high burden of kidney diseases and laid the groundwork for precision medicine.[48] The AASK trial aimed to evaluate the effects of antihypertensive drug class and BP targets on CKD progression in African American participants, finding that ramipril significantly slowed disease progression, particularly in those with proteinuria, demonstrating the

1950	Establishment of the National Institute of Arthritis and Metabolic Disease (NIAMD)
1972	Medicare ESRD program was initiated
1986	NIAMD was renamed as NIDDK
1983	DCCT Study: Benefits of intensive glycemic control on microvascular outcomes
1988	United States Renal Data System (USRDS) was established
1989	MDRD Study: Introduced MDRD equation for kidney function assessment
1995	HEMO Trial: Evaluated effects of dialysis dose and membrane flux
	AASK Trial: 1st large RCT of African American participants with CKD
2000	Establishment of the National Kidney Disease Education Program (NKDEP)
	DAC Studies: Evaluated effects of anti-platelet agents on arteriovenous access
2001	CRIC Study: Longitudinal prospective cohort study of adult CKD initiated
2003	ATN Trial: Joint initiative with VA to evaluate effects of intensive KRT in AKI
	CKiD Study: Longitudinal prospective cohort of pediatric CKD initiated
2006	HALT PKD: Evaluated BP and RAASi control on cyst growth in PKD
	FHN Trial: Evaluated frequency of HD on CV health and QOL
2009	CKD Biomarkers Consortium: Established for early detection of CKD
	ASSESS-AKI: Prospective longitudinal cohorts of adults and children with AKI
2010	SPRINT Trial: BP targets in CKD and cardiovascular disease risk
	NEPTUNE: Prospective longitudinal cohort GN (MCD, MN, FSGS)
2013	CureGN: Prospective longitudinal cohort of biopsy-proven GN (FSGS, MN, IgA, LN)
	TiME Trial: 1st pragmatic cluster RCT evaluating effects of dialysis duration
2017	KPMP launched: Molecular atlas of human kidney
2018	FDA approves SGLT2 inhibitors for CKD based on foundational NIDDK discoveries
2019	HOPE Trial: 1st major RCT with patient-reported outcome (pain interference) as primary end point in kidney disease–related research
2025	NIDDK 75th Anniversary

- Institutional/Foundational
- Chronic Kidney Disease (CKD)
- End-Stage Kidney Disease (ESKD)
- Acute Kidney Injury (AKI)
- Diabetes-Related Kidney Disease
- Genetic and Glomerular Disease
- Pediatric Nephrology

Figure 2. Chronology of selected NIDDK milestone studies. AASK, African American Study of Kidney Disease and Hypertension; ASSESS-AKI, Assessment, Serial Evaluation, and Subsequent Sequelae of AKI; ATN, Acute Renal Failure Trial Network; CKiD, Chronic Kidney Disease in Children; CRIC, Chronic Renal Insufficiency Cohort; CureGN, Cure Glomerulonephropathy; CV, cardiovascular disease; DAC, Dialysis Access Consortium; DCCT, Diabetes Control and Complications Trial; FDA, Food and Drug Administration; FHN, Frequent Hemodialysis Network; HALT PKD, Halt Progression of Polycystic Kidney Disease; HD, hemodialysis; HEMO, Hemodialysis; HOPE, HOPE Consortium Trial to Reduce Pain and Opioid Use in Hemodialysis; KPMP, Kidney Precision Medicine Project; LN, lupus nephritis; MCD, minimal change disease; MDRD, Modification of Diet in Renal Disease; MN, membranous nephropathy; NEPTUNE, Nephrotic Syndrome Study Network; NIAMD, National Institute of Arthritis and Metabolic Diseases; NKDEP, National Kidney Disease Education Program; QOL, quality of life; RAAS, renin-angiotensin-aldosterone system; RCT, randomized controlled trial; SGLT2, sodium-glucose cotransporter 2; SPRINT, Systolic Blood Pressure Intervention Trial; TiME, The Time to Reduce Mortality in End-Stage Renal Disease; USRDS, United States Renal Data System; VA, Department of Veterans Affairs.

superiority of angiotensin-converting enzyme inhibitors (ACEis) for kidney protection in this high-risk population. The NIDDK has also heavily invested in the Chronic Renal Insufficiency Cohort (CRIC) study (2001–ongoing), which remains one of the largest longitudinal CKD cohorts in the world and has served as a model for other longitudinal CKD studies around the globe.[49] Furthermore, the CRIC study has provided critical insights into CKD progression, cardiovascular risk, and prognostic biomarkers across diverse populations.[50–54] The CKD Biomarkers Consortium (2009–ongoing) has also been a seminal multidisciplinary collaborative research effort funded by the NIDDK that has advanced our understanding of biomarkers in kidney disease progression, risk prediction, and personalized

treatment for CKD, while also setting a precedent for how multi-institutional collaborations can be structured.[55,56] In addition, the Systolic Blood Pressure Intervention Trial (SPRINT) (2010–2015), which found that intensive BP control reduces cardiovascular risk leading to paradigm shifts in hypertension management and international practice guidelines, was primarily funded by the National Heart, Lung, and Blood Institute with additional support from the NIDDK, National Institute of Neurological Disorders and Stroke, and National Institute on Aging.[57–61]

Dialysis and Kidney Transplantation

The evolution of kidney failure care in the United States has been heavily influenced by research, infrastructure, and policy efforts supported by the NIDDK, which has aided in transforming kidney failure from a terminal condition into a chronic, manageable disease. In 1988, the NIDDK established the USRDS,[1,6] which has since become the world's most comprehensive nationwide registry of patients receiving KRT.[4] In 1999, the USRDS further expanded its scope to include comprehensive reports on CKD and pediatric kidney disease.[62] The USRDS has provided decades of epidemiologic data on the incidence, prevalence, modality trends, mortality, hospitalizations, prescription drug use, and costs of kidney disease care, which have influenced federal payment models, care benchmarks, and public health policies while serving as a vital resource for surveillance, policy development, and outcomes research.[11,12,63,64]

The NIDDK has also had a crucial role in supporting comparative effectiveness research to improve dialysis care. For example, the hemodialysis (HEMO) study (1995–2001) was a landmark randomized trial that evaluated the effects of dialysis dose and membrane flux in thrice-weekly in-center hemodialysis patients, which informed targets for dialysis dose and introduced more nuanced approaches to the hemodialysis prescription.[65] The Dialysis Access Consortium (DAC) studies (2000–2008), which evaluated the effects of clopidogrel on native arteriovenous fistulas (DAC AVF) and dipyridamole-aspirin on synthetic grafts, were the first large, multicenter trials specifically designed to prevent vascular access failure and shed light on the need for improved therapeutic strategies to improve fistula and graft maturation, respectively.[66,67] The Frequent Hemodialysis Network Daily and Nocturnal trials (2006–2010) were two major studies demonstrating that more frequent dialysis could improve physical function domains of quality of life in kidney failure, while also highlighting the practical trade-offs of intensive therapy.[68,69] The Time to Reduce Mortality in ESRD trial (TiME) (2013–2017) was a pragmatic, cluster-randomized controlled trial that was designed to test whether prescribing longer versus usual care length of dialysis sessions improved mortality, hospitalizations, and quality of life compared with usual care dialysis durations and has served as a milestone in trial methodology, showing that large-scale clinical trials can be integrated into real-world dialysis care by leveraging existing clinical infrastructure and electronic health records.[70] The Hemodialysis Fistula Maturation study (2010–2015) investigated the natural history of arteriovenous fistulas for hemodialysis and evaluated the effect of clinical characteristics, vascular ultrasound, vascular function, vein histology, and processes of care on arteriovenous fistula maturation.[71] More recently, the NIDDK-funded HOPE Consortium Trial to Reduce Pain and Opioid Use in Hemodialysis (2019–2024) is the first major trial addressing chronic pain, one of many burdensome symptoms experienced by patients undergoing long-term dialysis, with the primary outcome as a patient-reported outcome measure.[72–74]

AKI

The NIDDK has significantly advanced the prevention, earlier detection, targeted treatment, and understanding of the mechanisms of AKI through various landmark studies. In the early 2000s, NIDDK-funded researchers identified novel biomarkers such as neutrophil gelatinase–associated lipocalin (NGAL) and kidney injury molecule-1 (KIM-1), enabling earlier detection of kidney injury and improved monitoring of AKI.[75] The Acute Renal Failure Trial Network (ATN) study (2003–2008), jointly funded by the NIDDK and the Department of Veterans Affairs, was a multicenter trial that compared intensive (*i.e.*, more frequent or higher dose) versus less-intensive KRT strategies in critically ill AKI patients.[76] The ATN trial found that intensive therapy did not improve patient survival nor kidney recovery compared with standard dosing, challenging the prevailing belief that more aggressive KRT improves outcomes in AKI. The Assessment, Serial Evaluation, and Subsequent Sequelae of Acute Kidney Injury (ASSESS-AKI) study (2009–2020) was a multicenter prospective cohort study with longitudinal collection of biospecimens and serial evaluation of novel biomarkers and risk scores among adults and children with versus without AKI.[77–80] ASSESS-AKI has enabled characterization of the long-term sequelae of AKI, including its links to CKD progression, cardiovascular events, and mortality. In the past decade, the NIDDK has also spearheaded precision medicine in nephrology through initiatives including the Kidney Precision Medicine Project (KPMP) (2017–present), which aims to redefine AKI and CKD classification and treatment at the molecular level.[77,81–85] Using kidney biopsy specimens paired with single-cell RNA sequencing, proteomics, and imaging to identify cellular and molecular signatures of AKI, the KPMP aims to create a detailed molecular atlas of human kidney tissue from patients with AKI and CKD to better distinguish different kidney disease subtypes based on pathophysiology that will inform tailored treatment approaches. The NIDDK has also invested in the (Re)Building a Kidney consortium, which has convened experts in bioengineering, developmental biology, kidney physiology, stem cell research and regeneration, and kidney injury to elucidate both endogenous kidney repair mechanisms and the generation of new kidney tissue.[86]

DKD

In parallel, foundational research and landmark studies supported by the NIDDK have deepened understanding of the mechanisms and management of DKD. The Diabetes Control and Complications Trial (DCCT) (1983–1993) and its follow-up Epidemiology of Diabetes Interventions and Complications (EDIC) study (1994–ongoing) were both funded by the NIDDK and have been two of the most influential studies in type 1 diabetes, showing that early intensive glycemic control reduces long-term microvascular

complications including DKD, and establishing the concept of "glycemic memory."[87,88] In addition, the United Kingdom Prospective Diabetes Study (UKPDS) (1977–1997) was partly supported by NIDDK and similarly showed that early intensive therapy confers long-term reduction of microvascular complications in type 2 diabetes.[89,90] The Family Investigation of Nephropathy and Diabetes (FIND) study (1999–2006) was a pioneering multicenter investigative effort that used family- and population-based genetic methods across diverse populations to identify key genetic variants as risk loci for diabetic nephropathy.[91] In addition to establishing a robust biorepository and advancing understanding of genetic risk in DKD, FIND data contributed to the discovery and confirmation of the *APOL1* risk variants that contribute to the high rates of kidney diseases in individuals of West African descent.

The NIDDK was also instrumental in supporting investigators who laid the scientific and translational groundwork leading to the development and clinical use of ACEis and angiotensin receptor blockers (ARBs) as cornerstone therapies for DKD[92–100] and other kidney diseases (*e.g.*, hypertensive CKD[48] and PKD[101,102]). Although the pivotal clinical trials of these agents were industry-sponsored (*i.e.*, Collaborative Study Group Captopril Trial,[103] Irbesartan Diabetic Nephropathy Trial [IDNT],[104] and Reduction of Endpoints in Non-Insulin-Dependent Diabetes Mellitus with the Angiotensin II Antagonist Losartan [RENAAL][105]), the NIDDK's investment in researchers as well as individual studies focused on renin-angiotensin-aldosterone system (RAAS) biology, DKD pathophysiology, disease modeling, biomarker development, and natural history has enabled the successful development and clinical validation of ACEis and angiotensin receptor blockers for DKD.[92–100,106–110] Foundational research leading to the discovery and cloning of SGLT2 was supported by the NIDDK[1,6,13–16] and led to industry-funded trials (*i.e.*, Canagliflozin and Renal Events in Diabetes with Established Nephropathy Clinical Evaluation [CREDENCE],[111] Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease [DAPA-CKD],[112] Empagliflozin and Progression of Kidney Disease in CKD Patients [EMPA-Kidney][113]) that have demonstrated the cardio-renal benefits of SGLT2 inhibitors in diabetic and nondiabetic CKD patients. Notably, in 2024, NIDDK grant recipients Dr. Joel Habener and Dr. Svetlana Mojsov were honored with the Lasker–DeBakey Clinical Medical Research Award along with another scientist for their role in the discovery and development of glucagon-like peptide-1–based drugs[1,6] that have demonstrated cardiorenal benefits in DKD and revolutionized the treatment of obesity.

Genetic Kidney Diseases and Glomerular Diseases

NIDDK-funded research has informed the hereditary nature, natural progression, and management of diseases such as PKD. Studies of the *PKD1* and *PKD2* mutations[114–116] and the role of cyclic AMP in PKD[24,117–121] were funded by the NIDDK and have enabled molecular diagnostics[122,123] and targeted treatments including tolvaptan.[33,34] The Consortium for Radiologic Imaging Studies of Polycystic Kidney Disease (CRISP I, II, and III; 2000–2015)[124] established total kidney volume as a noninvasive imaging biomarker for autosomal dominant PKD (ADPKD) progression, enabling its use as a surrogate end point in major PKD trials. The Halt Progression of Polycystic Kidney Disease (HALT-PKD) A and B trials (2006–2014), which evaluated BP control and combined RAAS inhibition on cyst growth in ADPKD, found that intensive BP slowed ADPKD progression and eGFR decline, whereas no significant benefit was found for dual RAAS blockade versus ACEi alone.[102,125–127] It also bears mention that the NIDDK has funded the Polycystic Kidney Disease Research Resource Consortium that provides ongoing coordinated support and resources for PKD investigations throughout the United States that are laying the groundwork for future discoveries of additional therapeutic interventions.[128–133]

The discovery of *APOL1* genetic risk variants linked to higher risk of FSGS and hypertensive kidney disease in populations of West African ancestry by Dr. Pollak's team was also driven by NIDDK support.[134,135] In addition to providing critical insights into the genetic underpinnings of the heightened burden of CKD in individuals of West African ancestry, this seminal research has prompted development of genetic risk screening/stratification and personalized treatment approaches. Furthermore, discovery of the *APOL1* gene has paved the way for the development of precision pharmacotherapeutic strategies targeting *APOL1*-associated kidney disease.[136] These studies have also informed the NIDDK-funded Nephrotic Syndrome Study Network (NEPTUNE) Consortium (2010–ongoing)[137–141] and Cure Glomerulonephropathy Network (CureGN) (2013–ongoing), two multicenter longitudinal cohorts of adults and children with biopsy-proven glomerular diseases (*e.g.*, FSGS, membranous nephropathy, minimal change disease, and IgA nephropathy).[140,142–147] By integrating clinical, genetic, transcriptomic, proteomic, histopathologic, and imaging data, NEPTUNE and CureGN have made fundamental advances in the field by enabling *APOL1* genetic research in glomerular diseases, accelerating biomarker discovery, and shifting nephrology toward molecular diagnostics and precision therapy. Following Dr. David Salant's team's identification of the phospholipase A2 receptor (PLA2R) as the primary antigen in primary membranous nephropathy,[148] the NIDDK provided funding support to validate and expand this discovery toward the development of reliable antibody assays, as well as research linking PLA2R autoantibody levels to clinical outcomes. Furthermore, the NEPTUNE and CureGN research networks played a key role in validating PLA2R as a biomarker across large, multicenter, biopsy-confirmed cohorts, as well as confirming that PLA2R positivity is associated with active disease and that declining titers precede remission.[148,149]

Pediatric Nephrology

The NIDDK has also had a major effect on pediatric nephrology through investment in large-scale research consortia and longitudinal studies. In addition to the aforementioned ASSESS-AKI, NEPTUNE, and CureGN studies, which included children with CKD, the NIDDK has funded the Chronic Kidney Disease in Children (CKiD) study (2003–ongoing), a longitudinal prospective cohort of children with CKD that had aided in informing the natural history and progression of CKD in children; identifying clinical, demographic, genetic, and biochemical risk factors

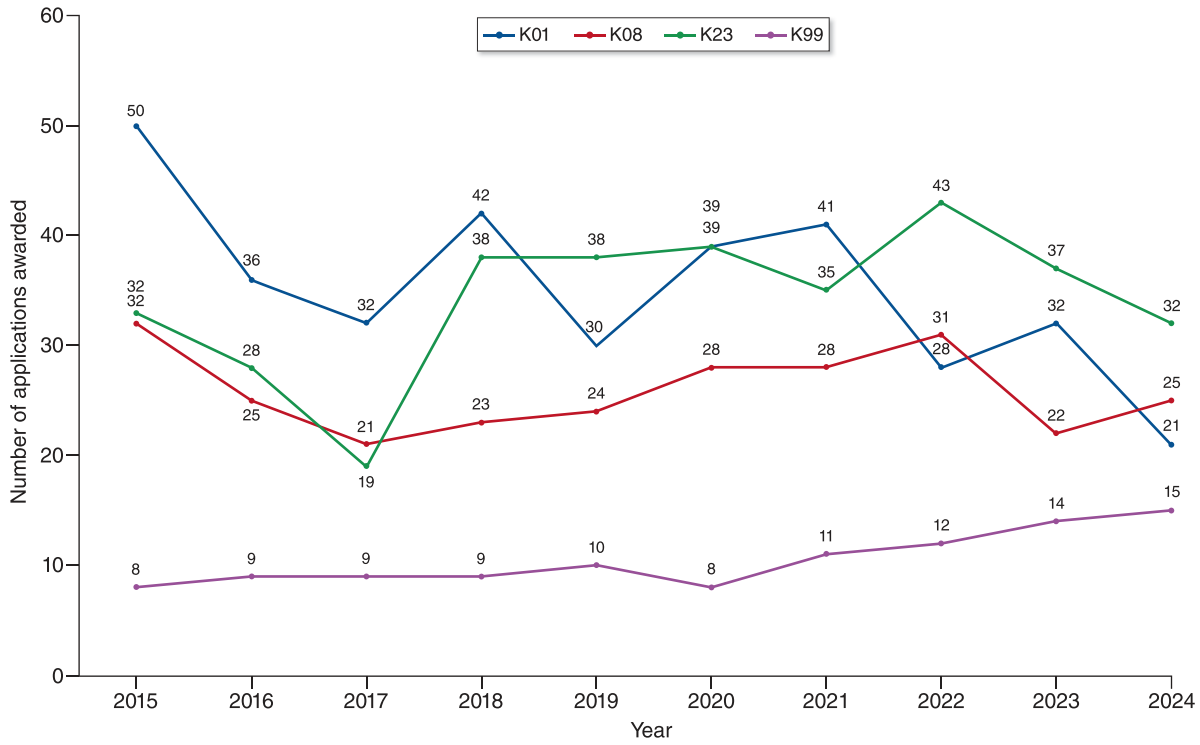

Figure 3. Number of NIDDK training grants awarded from 2015 to 2024.

for kidney disease progression; and evaluating the effect of CKD on children's growth and development, cardiovascular health, quality of life, and cognitive and neurobehavioral function.[150–153] The Treatment Options for Type 2 Diabetes in Adolescents and Youth (TODAY) trial (2004–2011) and its long-term follow-up TODAY2 studies (2011–2020) have been landmark NIDDK initiatives that have expanded understanding of youth-onset type 2 diabetes. Although the TODAY study was the first large-scale trial to define optimal treatment in youth-onset type 2 diabetes and demonstrate the need for early combination therapy, TODAY2 has provided evidence that youth-onset type 2 diabetes is more aggressive than adult-onset diabetes with respect to early complications, including critical insights into characterizing the early development and progression of kidney disease in this population.[154–156]

NIDDK Central Repository–Resources for Research

Many of the above trials and prospective cohorts have also contributed to a unique research repository and infrastructure, known as the NIDDK Central Repository–Resources for Research (NIDDK-CR R4R). The NIDDK-CR R4R provides researchers with access to banked clinical data and biologic specimens and enables current and future generations of investigators to explore novel hypotheses and drive new scientific discoveries.

Training, Career Development, and Mentorship

The NIDDK has also been committed to investing in the training, development, and mentorship of future kidney health leaders, including scientists, clinicians, and educators, across all career stages.[157–160] For example, the NIDDK has strategically prioritized support for early career investigators, recognizing that sustained mentorship and funding during the fragile transition from training to independence are essential to long-term success. As such, the NIDDK provides a suite of career development mechanisms (Figure 3), including K01 (Mentored Research Scientist Development), K08 (Mentored Clinical Scientist Development), and K23 (Mentored Patient-Oriented Research Career Development) awards, which provide protected research time and mentorship to early career investigators across the kidney research continuum. These awards have enabled hundreds of emerging nephrologists, clinicians, epidemiologists, and basic scientists to launch impactful careers. Recognizing the challenges of securing initial R01 funding, the NIDDK has also provided bridge support programs that assist promising investigators with high-quality but unfunded R01 applications, reducing attrition from academic research careers. Early Independence Awards and R56 mechanisms have similarly enabled researchers to sustain momentum during these critical early transitions. At even earlier stages, the NIDDK has offered individual F31 (predoctoral) and F32 (postdoctoral) fellowships to support trainees pursuing research focused on kidney health and diseases, allowing them to build publication records and develop grant/scientific writing and analytical skills. The NIDDK has also funded numerous Loan Repayment Awards,[161] a federal initiative that provides student loan forgiveness for eligible researchers who are committed to conducting biomedical or behavioral

research; the Loan Repayment award has been an important tool for recruiting and retaining scientists in research careers by easing the burden of educational debt.

The NIDDK has also provided support to academic and medical institutions to develop high-quality, structured training environments. For example, the T35 (Ruth L. Kirschstein National Research Service Award Short-Term Institutional Research Training Grant) is a short-term institutional training grant designed to provide intensive, mentored research experiences for students in health professional degree programs during the summer or other breaks from their academic curriculum. In prior years, the NIDDK also offered the T32 (Ruth L. Kirschstein National Research Service Award Institutional Research Training Grant) to fund comprehensive training programs for postdoctoral fellows at individual institutions. More recently, the Division of Kidney, Urologic, and Hematologic Diseases (KUH) of the NIDDK has reshaped institutional training awards in creating the KUH Research Training Network, which consists of Institutional Network Awards (U2C-TL1). The U2C cooperative component coordinates and provides administrative support for activities across multiple cores (*i.e.*, administrative, professional development, and training), while the TL1 training component allows the institution to recruit predoctoral and postdoctoral trainees conducting KUH-focused research. Many of these T-series and F-series trainees have gone on to receive K- and R-series awards, highlighting the success of NIDDK's training programs.

Advancing Equal Access to Kidney Disease Care

Through the years, the NIDDK has supported, and continues to support, improving kidney health for all Americans, as well as building a scientific workforce that promotes this goal. To this end, the institute has implemented multifaceted approaches to ensure all communities have equal access to health care and participation in research. First, the NIDDK has demonstrated leadership in building broad representation in the kidney research workforce.[158,159,162] Through various funding mechanisms, such as R25 education and training programs and KUH Predoctoral to Postdoctoral Fellow Transition (F99/K00) awards, the institute has supported equal access to research vis-à-vis its support of investigators from multiple backgrounds and viewpoints.

Epidemiologic studies supported by the NIDDK and from the USRDS have also uncovered gaps experienced by communities with the highest disease burden, as well as the role of social determinants of health, rural residence, and limited access to specialty care on kidney health outcomes.[163–166] NIDDK-supported research has also explored the complex interplay between genetics (*e.g.*, *APOL1* variants) and social context, demonstrating that both biologic and structural factors must be addressed to achieve optimal outcomes. Furthermore, the institute has also sought to ensure that kidney disease research reflects all Americans, and many of its major trials (*e.g.*, AASK, CRIC, TODAY/TODAY2, and SPRINT) were designed to have a broad representation. Major NIDDK-supported studies such as NEPTUNE, CureGN, KPMP, and HOPE trial have also incorporated patient advisory panels to ensure that research priorities reflect the needs and voices of people living with kidney diseases.

To reach underresourced communities and promote early detection and prevention, NIDDK has also promoted public health and community engagement initiatives.[167] For example, the NIDDK established the National Kidney Disease Education Program (NKDEP) (2000–2019), which promoted evidence-based interventions to improve understanding, detection, and management of kidney diseases, including identification of high-needs patients.[168,169] While the National Kidney Disease Education Program's initial focus was on standardizing CKD screening, it continued to expand its scope toward public health outreach by emphasizing improved access to screening, awareness, education, and early detection and referral in communities with limited access; supporting and developing patient-centered education; promoting community-based partnerships; and delivering kidney health education in the communities where patients reside (barbershop and church outreach, health coalitions, and through community health workers).

Collaborative Partnerships

Since its inception, the NIDDK has recognized that scientific discovery alone is insufficient to improve population health outcomes. To achieve meaningful progress in kidney disease prevention, detection, and treatment, the institute has built and nurtured a wide network of partnerships that span academic centers, federal agencies, patient advocacy organizations, nonprofits, industry stakeholders, and community groups (Figure 4). These NIDDK partnerships to align national efforts in kidney health have included collaborations with the Centers for Medicare and Medicaid Services, Veterans Health Administration, Patient-Centered Outcomes Research Institute, Centers for Disease Control and Prevention, ASN, and National Kidney Foundation. For example, ASN and NIDDK have partnered together in advancing nephrology workforce development; advocating for kidney research funding to Congress; sponsoring joint symposia, workshops, and conferences; providing feedback on initiatives impacting the kidney research community; and supporting efforts to enhance patient involvement in the research process. In addition to joint research initiatives such as KPMP,[170,171] ASN and NIDDK have established private-public partnerships such as the Kidney Health Initiative[172] that was cofounded in 2012 with FDA involvement to accelerate innovation in kidney disease treatment, research tools, and patient engagement spanning areas related to patient-reported outcomes, clinical trial end points, kidney replacement device innovation, and wearable dialysis. The NIDDK has also convened multidisciplinary experts through workshops and symposia on key topics including cellular physiology, the gut microbiota–kidney axis, precision medicine trials, kidney function assessment, and postdialysis fatigue, to foster collaboration, share emerging insights, and guide future research. NIDDK-funded research has also facilitated subsequent industry-sponsored clinical trials that have led to the development of pharmacologic and therapeutic

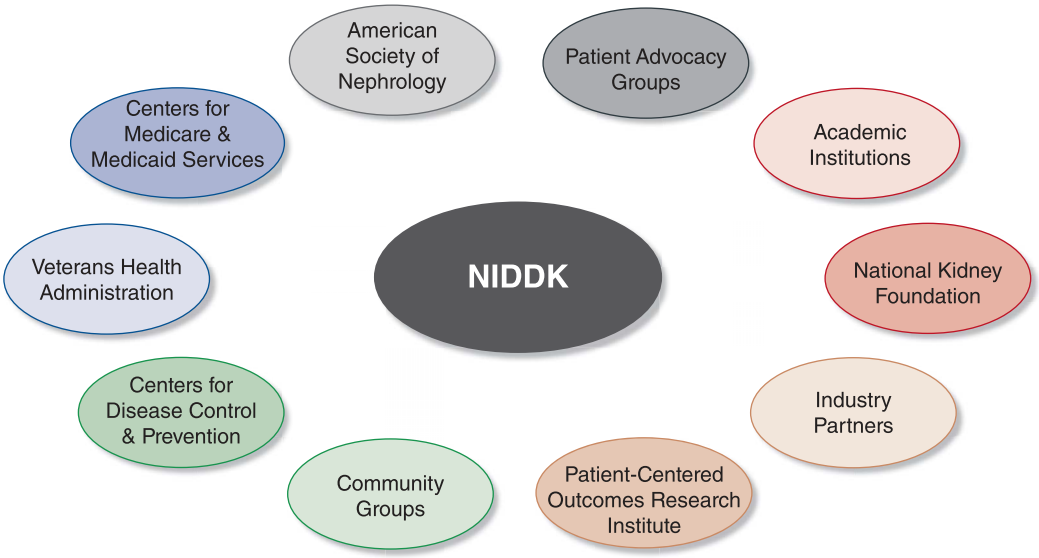

Figure 4. NIDDK's collaborative partnerships in kidney health.

interventions aimed at slowing the progression of kidney diseases, delaying the need for dialysis or transplantation, reducing morbidity and mortality, and enhancing patient quality of life. These collaborations have strengthened the reach of NIDDK-supported science and have enhanced its real-world impact.

Conclusion

Over the course of its 75-year history, the NIDDK has demonstrated a steadfast commitment to advancing kidney research and improving patient outcomes. Through its investment in landmark discoveries, rigorous bench and clinical investigations, and technologic innovations, the NIDDK has been the scientific engine driving improved understanding, prevention, and treatment of kidney diseases worldwide. These accomplishments have translated into enhanced diagnostic approaches, evidence-based therapeutic strategies, and improved standards of care, benefiting all Americans. These foundational investments by the NIDDK are also positioned to generate long-term advancements in the field of nephrology. For example, embracing the omics revolution—encompassing genomics, proteomics, metabolomics, and beyond—and enabling the large-scale integration of molecular and clinical data hold the potential to accelerate translational research, enhance disease stratification, and drive the development of targeted therapies for kidney diseases. The success of research entities such as the NIDDK also underscores the critical and evolving need to modernize the biomedical research infrastructure in the United States. As scientific advances accelerate, particularly in fields such as precision medicine, data integration, and multiomics, the current infrastructure must evolve to support the scale, complexity, and collaborative demands of contemporary research. Modernizing data systems, enhancing interoperability across institutions, expanding biobanking and computational resources, and fostering cross-disciplinary partnerships are essential steps to fully capitalize on the momentum generated by agencies like the NIDDK and to sustain innovation in nephrology and related disciplines.

In reflecting on the NIDDK's remarkable achievements through the years, it is also important to recognize the enormous dedication of its leadership and staff, as well as the strength of its strategic efforts in collaborating with extramural researchers, health care professionals, patients, professional societies, and policymakers. Indeed, the NIDDK and the institutes that comprise the NIH have been a leading force in establishing the United States as a global leader in biomedical research and innovation and have brought us significantly closer to a future free of kidney diseases.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at http://links.lww.com/JSN/F458.

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